

EXPERIMENT 5

(A)

(A) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

If a user has never received any confirmation request, their confirmation rate should be considered **0**.

Display the user ID along with the confirmation rate rounded to **2 decimal places**. The result can be returned in any order.

CODE:

```
SELECT T.REQUEST_AT AS DAY,
       ROUND(
         AVG(
           CASE
             WHEN T.STATUS IN
              (
                'cancelled_by_driver',
                'cancelled_by_client'
              )
             THEN 1
             ELSE 0
           )
        )
```

),2)

FROM Trips t

JOIN Users c

ON

t.client_id = c.users_id

JOIN Users d

ON

t.driver_id = d.users_id

WHERE c.banned = 'No'

AND d.banned = 'No'

AND t.request_at

BETWEEN '2013-10-01' AND '2013-10-03'

GROUP BY T.REQUEST_AT;

	user_id [PK] integer	confirmation_rate numeric
1	2	0.50
2	3	0.00
3	6	0.00
4	7	1.00

(B) A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between **2013-10-01** and **2013-10-03**, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

CODE:

SELECT

t.request_at AS day,

```

ROUND(
    SUM(CASE WHEN t.status LIKE 'cancelled%' THEN 1 ELSE 0 END)::NUMERIC /
    COUNT(*),
    2
) AS confirmation_rate
FROM Trips t
JOIN
    Users c ON t.client_id = c.users_id AND c.banned = 'No'
JOIN
    Users d ON t.driver_id = d.users_id AND d.banned = 'No'
WHERE
    t.request_at BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY
    t.request_at
ORDER BY day;
  
```

	day date	confirmation_rate numeric
1	2013-10-01	0.33
2	2013-10-02	0.00
3	2013-10-03	0.50